RESEARCH ARTICLE · EDUCATION · ARTIFICIAL INTELLIGENCE · STUDENT WELL-BEING

The Impact of Generative AI Tool Usage on Student Academic Performance, Skill Retention, and Burnout Risk

A cross-sectional analysis of 50,000 students

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October 2026

50,000
university students

16
variables, no missing data

7
pre-specified hypotheses

ARTICLE AT A GLANCE

Four numbers that frame the study

87.5%

of students improved their GPA after AI-assisted learning

+0.203

mean GPA change across 50,000 students

83.1%

high-burnout rate at 25–40 AI hours per week

+11

points of skill-retention edge for advanced prompt engineers

> Design: cross-sectional · N = 50,000 · 16 variables · no missing data. Companion tool: an interactive Streamlit dashboard.

ROADMAP

What this presentation covers

01 Background
Literature, theory and the evidence gap

02 Objectives and hypotheses
Five objectives, seven hypotheses

03 Methodology
Sample, measures, models, ethics

04 Results
Descriptives, correlations, thresholds, ML models

05 Discussion
Mechanisms, recommendations, limitations

06 Conclusions
Six takeaways and future research

07 Appendices
Codebook, post-hoc tests, diagnostics

+ Companion dashboard
Five-module Streamlit application

THE CENTRAL QUESTION

Not whether students should use AI, but how much, how well, and under what rules.

The study moves past “does AI help or hurt?” toward usage intensity, skill level and institutional context.

01

Background

Why generative AI in education needs large-scale evidence, and the theory that guides the hypotheses.

CONTEXT . RESEARCH GAP

A fast-moving technology with a thin evidence base

ADOPTION

Rapid adoption

ChatGPT arrived in November 2022, followed by Claude, GPT-4 and domain-specific models, reshaping academic work.

EVIDENCE

Fragmented evidence

Findings are contradictory and rest on small institutional studies that lack statistical rigour.

POLICY

Policy by intuition

Institutions range from strict bans to citation-based permission, often without outcome data.

What this study adds

BEFORE

The gap

Differences in student outcomes across policy environments are largely undocumented at scale, and usage patterns are rarely linked to several outcomes at once.

THIS STUDY

The contribution

50,000 students across disciplines, academic levels and institutions. Three outcomes: GPA, skill retention and burnout risk. Contextual moderators: prompt skill, policy, study hours, tool diversity.

> Hypothesis behind the design: the policy debate has focused on the wrong variable.

Generative AI differs from earlier ed-tech

d = 0.40

moderate effect of intelligent tutoring systems in earlier meta-analyses

Kulik and Fletcher, 2016; VanLehn, 2011

Constrained versus open-ended

Earlier systems used predetermined solution paths. LLMs co-construct outputs, so transfer of past findings is speculative.

Content retrieval

Tutoring

Co-creation

Assessment

Adaptive learning

> Kasneci et al. (2023) taxonomy: generative AI maps most directly onto co-creation.

LITERATURE · BURNOUT · SKILL RETENTION

Burnout is common and non-linear in workload

20–50%

burnout prevalence in higher-education samples, depending on measurement

Schaufeli et al., 2002; Marčinko et al., 2020

Job demands–resources

Burnout follows a chronic imbalance between demands and personal resources (Bakker and Demerouti, 2007).

Technostress

Technology can raise subjective strain even while reducing objective workload (Brod, 1982).

Yerkes–Dodson logic predicts an optimal moderate zone with strain rising at both extremes.

Struggle builds retention, scaffolding builds understanding

RISK

Desirable difficulties

Spacing, interleaving and productive struggle improve long-term retention and transfer (Bjork and Bjork, 1992). Tools that compress steps may remove that struggle.

OPPORTUNITY

Cognitive apprenticeship

Guided, modelled problem-solving makes expert reasoning explicit and can strengthen learning (Collins et al., 1989).

Prediction: direct answers should hurt retention; iterative, scaffolded use should protect it.

Prohibition, permission and harm reduction

BAN

Prohibition

Argues AI use breaches academic-integrity norms and outsources cognition (Bruckner and Hensley, 2023).

ENCOURAGE

AI literacy

Argues literacy is a core competency and prohibition only displaces use (Mollick and Mollick, 2023).

CITATION

Harm reduction

Borrowed from public health: channel inevitable use into lower-risk patterns with disclosure (Zinberg, 1984).

Maps onto three policies tested: Strict Ban, Allowed with Citation, Actively Encouraged.

Skill may matter more than the number of tools

SKILL

Prompt engineering

Problem decomposition, task specification, scaffolding, iterative refinement and evaluation of intermediate outputs. These mirror metacognitive skills (Wei et al., 2023; Zamfirescu-Pereira et al., 2023).

LOAD

Tool proliferation

Cognitive load theory (Sweller, 1988) fuels an institutional worry that using many AI platforms overloads working memory.

Tested directly: H5 on prompt skill and H2 on tool diversity.

THEORETICAL MAP

Seven lenses, one set of predictions

Lens	Core idea	Predicts in this study
Yerkes–Dodson	Moderate arousal is optimal	Inverted-U for usage hours
Job demands–resources	Strain from demand–resource imbalance	Burnout rises with heavy use
Technostress	Tools raise subjective strain	Heavy use adds psychological load
Desirable difficulties	Struggle aids retention	Direct answers lower retention
Cognitive apprenticeship	Guided reasoning teaches	Scaffolded use protects retention
Self-determination	Autonomy sustains well-being	A dependency threshold
Harm reduction	Channel use, do not ban it	Citation beats strict bans

02

Objectives and hypotheses

Five objectives and seven testable predictions.

STUDY OBJECTIVES

Five questions the analysis answers

1. **Characterise usage and performance**
   Relate generative AI usage to academic performance in 50,000 students.

2. **Test non-linearity**
   Is there an optimal usage window beyond which strain escalates?

3. **Rank the predictors**
   Usage volume versus tool diversity versus skill level for each outcome.

4. **Compare policy frameworks**
   Strict Ban, Allowed with Citation and Actively Encouraged.

5. **Test the skill multiplier**
   Does prompt proficiency turn AI from shortcut into pedagogical tool?

HYPOTHESES

Usage shape, burnout, protection and dependency

H1

Inverted-U usage

Moderate weekly AI use goes with the best GPA and retention; very high use shows diminishing or negative returns.

H2

Hours drive burnout

Weekly AI hours is the strongest burnout correlate ($|r| \geq 0.30$); tool diversity is negligible ($|r| < 0.10$).

H3

Offline study protects

Traditional study hours relate negatively to burnout, also after controlling for AI hours.

H4

Dependency threshold

Perceived dependency has a non-linear link to retention and burnout, with a threshold where outcomes worsen sharply.

Prompt skill, policy and use case

H5

Skill moderates learning

Advanced prompt users retain more than beginners, with a smaller or negligible GPA difference.

H6

Bans backfire

Strict bans: higher burnout and lower retention, but no change in perceived dependency.

H7

Use case matters

Scaffolding uses (debugging, ideation, summarising) beat direct answer generation on retention.

03

Methodology

Design, measures, statistical strategy and ethics.

STUDY DESIGN

Cross-sectional, large and complete

50,000
university students across institutions and disciplines

16
measured variables per student

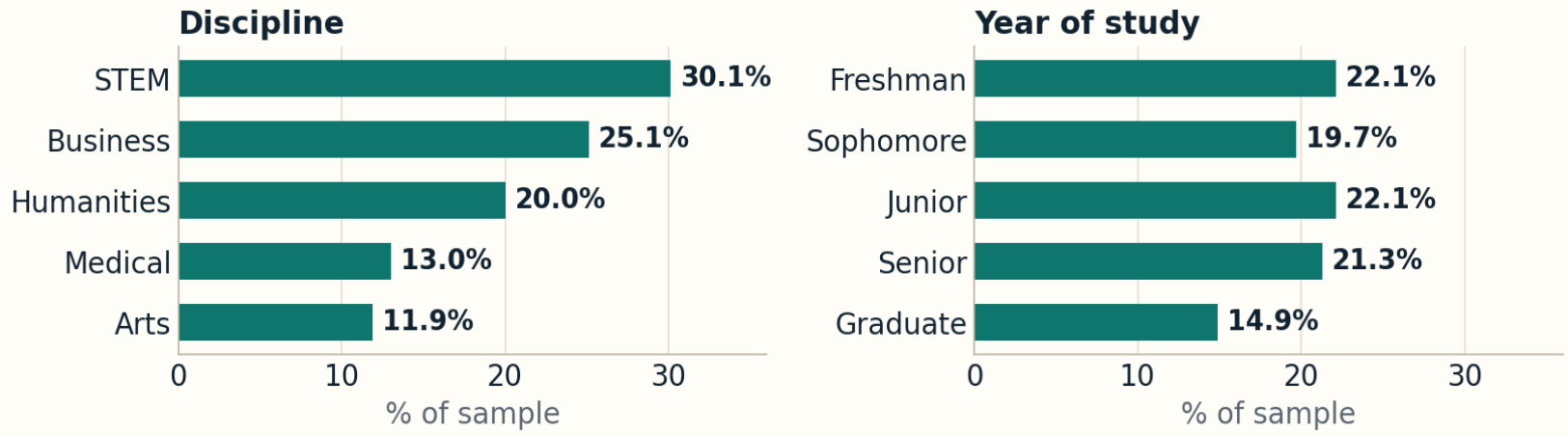
100%
data completeness, no imputation needed

1
academic term of data collection

> Single-term data allow precise description of concurrent associations but limit causal inference.

SAMPLE COMPOSITION

Balanced across disciplines and years



> AI subscriptions: 57.7% free, 42.3% paid. Policy groups: 30% Strict Ban, 36% Citation, 34% Encouraged.

Three outcomes capture performance, learning and strain

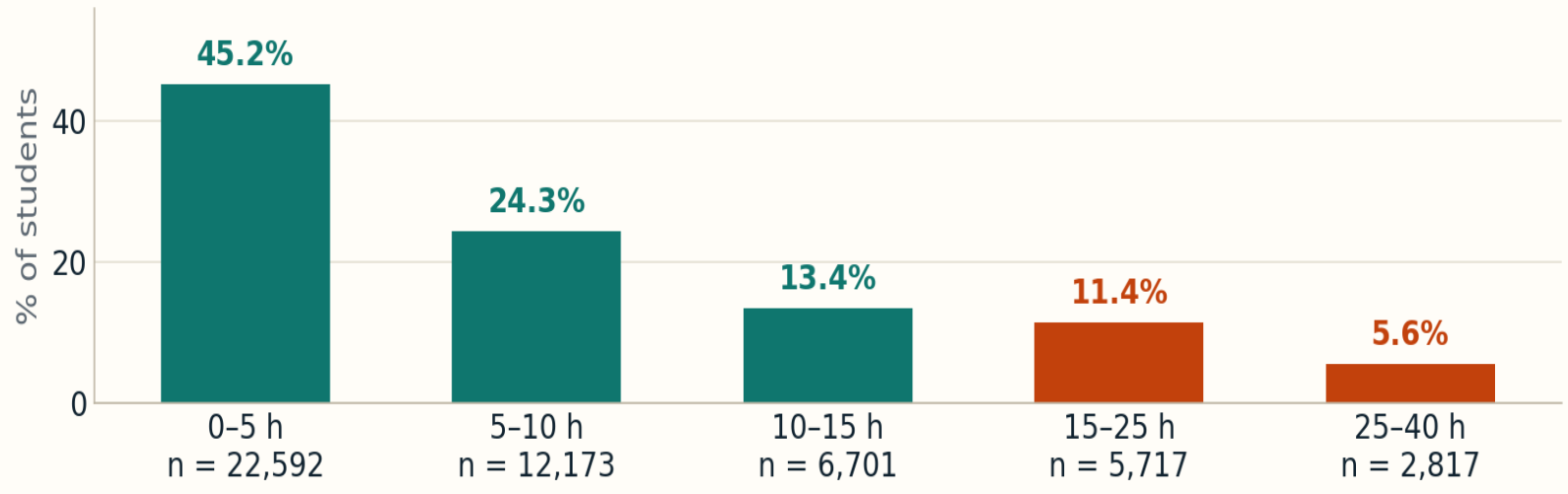
Outcome	How it was measured	Sample profile
Academic performance	Post-semester GPA on a 4.0 scale; change = post minus pre GPA	Mean GPA 3.349; change +0.203
Skill retention	Standardised 0–100 assessment of material recalled and applied one week after instruction	Mean 75.8 (SD 13.3)
Burnout risk	Low, Medium or High from validated symptom scales (exhaustion, cynicism, efficacy); continuous score also available	Low 32.74%, Medium 42.29%, High 24.97%

Seven predictors of interest

Variable	Scale	Descriptor
Weekly AI hours	Hours per week	Self-report plus platform analytics
Perceived AI dependency	1–10	Mean 3.5 (SD 1.8); threshold at 6/10
Prompt engineering skill	Beginner, Intermediate, Advanced	Demonstrated tasks plus self-report
Tool diversity	1 to 5+ tools	Mean 2.80 (SD 1.19)
Traditional study hours	0–40 per week	Offline study time
Institutional policy	Three frameworks	Strict Ban, Citation, Encouraged
Primary use case	Five categories	Shortcut versus scaffolding

USAGE BANDS . GROUPING VARIABLES

Most students use AI lightly; a small tail uses it heavily



17.1% of students report more than 15 hours a week; 5.6% report more than 25.

Near-equal groups keep comparisons clean

Institutional policy
Strict Ban 15,000
Allowed with Citation 18,000
Actively Encouraged 17,000

Prompt skill
Beginner 16,750
Intermediate 16,750
Advanced 16,500

Primary use case
Debugging 24.59%
Copywriting 24.02%
Ideation 21.44%
Summarising 17.27%
Direct answers 12.68%

Balanced prompt-skill groups remove unequal group size as an explanation for later contrasts.

STATISTICAL ANALYSIS

Four complementary analytic strands

ASSOCIATION

Correlation

Pearson and Spearman, two-tailed $p < 0.001$. Effect sizes per Cohen: small 0.10–0.29, medium 0.30–0.49, large ≥ 0.50.

SHAPE

Regression

Multiple linear models, usage² term, splines with knots at 5, 15 and 25 hours, and piecewise slopes.

PREDICTION

Machine learning

Gradient Boosting, Random Forest and Linear Regression for GPA and retention; Logistic Regression and Random Forest for burnout class.

SUBGROUPS

Stratified tests

ANOVA with Tukey HSD across discipline, year of study and policy; interaction terms for policy.

ROBUSTNESS

Normality failed, so the analysis used several methods

Variable	K^2	Decision
GPA change	1,264.65	Reject
Weekly AI hours	14,386.57	Reject
Study hours	411.56	Reject
Skill retention	500.58	Reject

Parametric and rank-based
Pearson alongside Spearman correlations.

Assumption-free models
Machine learning needs no distributional assumptions.

Flexible shape
Polynomial and piecewise regression.

All D’Agostino–Pearson tests gave $p < 0.001$. Residuals, the real requirement, satisfied normality.

ANALYTIC WORKFLOW . ETHICS AND DATA PRIVACY

Four stages from raw data to a prediction interface

STAGE 1
Audit

Profile 16 variables, check completeness, ranges and normality.

STAGE 2
Infer

Correlations, paired GPA test, ANOVA and Tukey HSD post-hoc tests.

STAGE 3
Model

Train supervised models on a held-out test set of 10,000 students (20%).

STAGE 4
Translate

Turn model outputs into an interactive interface and stakeholder advice.

> Every algorithm is reported side by side, so readers can judge whether gains come from model complexity or from signal.

De-identified, aggregated, STROBE-aligned

De-identified data

Names, specific course enrolment and institutional identifiers were removed before analysis.

Aggregated records

Individual students cannot be identified from the analytic dataset.

Reporting standard

Follows the STROBE guidelines for cross-sectional observational studies.

> Ethical oversight statement in the manuscript: [IRB protocol number or description of ethical oversight].

04

Results

From descriptives to thresholds, moderators and prediction models.

DESCRIPTIVE FINDINGS

The analytic sample at a glance

3.146
mean pre-semester GPA (SD 0.479)

3.349
mean post-semester GPA (SD 0.496)

+0.203
mean GPA change

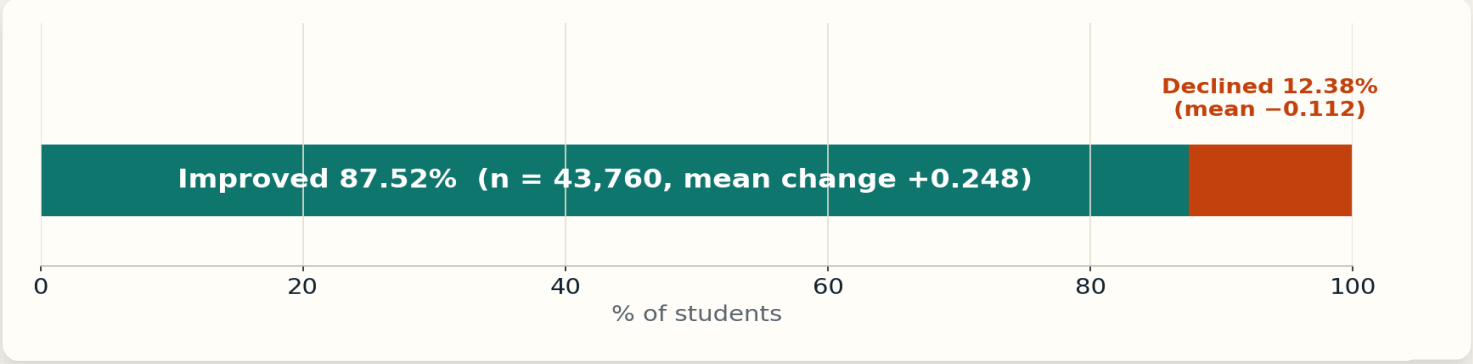
75.8
mean skill retention (SD 13.3)

3.5
mean perceived dependency, 1–10 (SD 1.8)

2.80
mean tools per student (SD 1.19)

ACADEMIC PERFORMANCE

87.5% of students improved their GPA



Mean change +0.203 (95% CI (95% CI 0.197–0.210), paired t paired t = −242.7, p < 0.001. 0.001. Unchanged: 0.10%.

BURNOUT RISK DISTRIBUTION

One in four students is at high risk

32.74%
Low risk
n = 16,370

42.29%
Medium risk
n = 21,143

24.97%
High risk
n = 12,487

> Burnout is not spread randomly across usage bands: $\chi^2 = 5{,}847.3$, $p < 0.001$.

DISTRIBUTIONS

Heavy AI users are a numerically small group

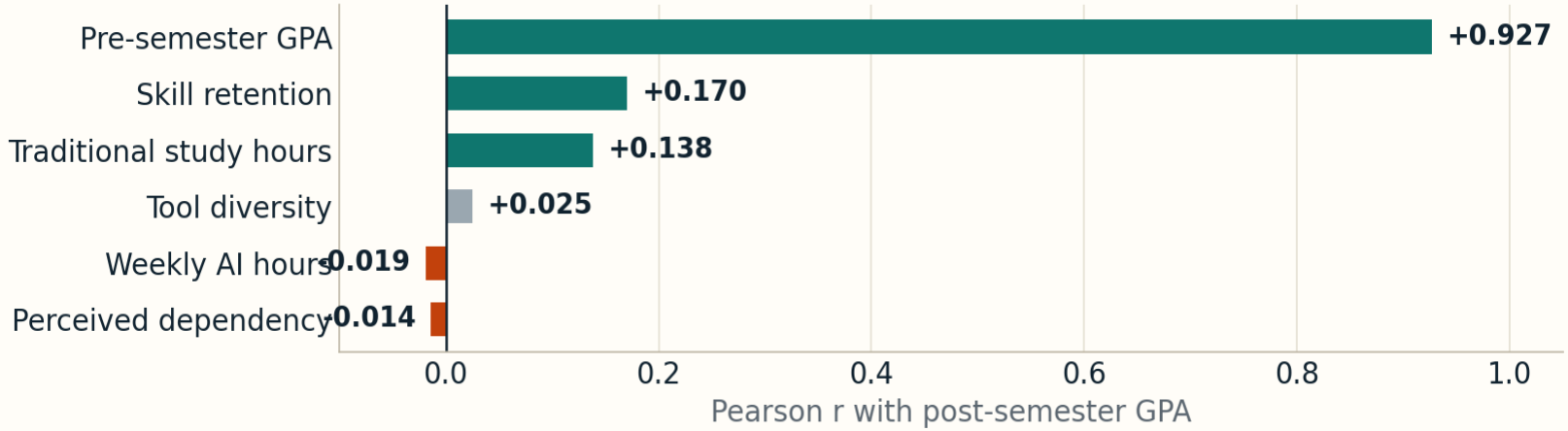
17.1%
of students report more than 15 hours a week

5.6%
report more than 25 hours a week

0.176
coefficient of variation in skill retention

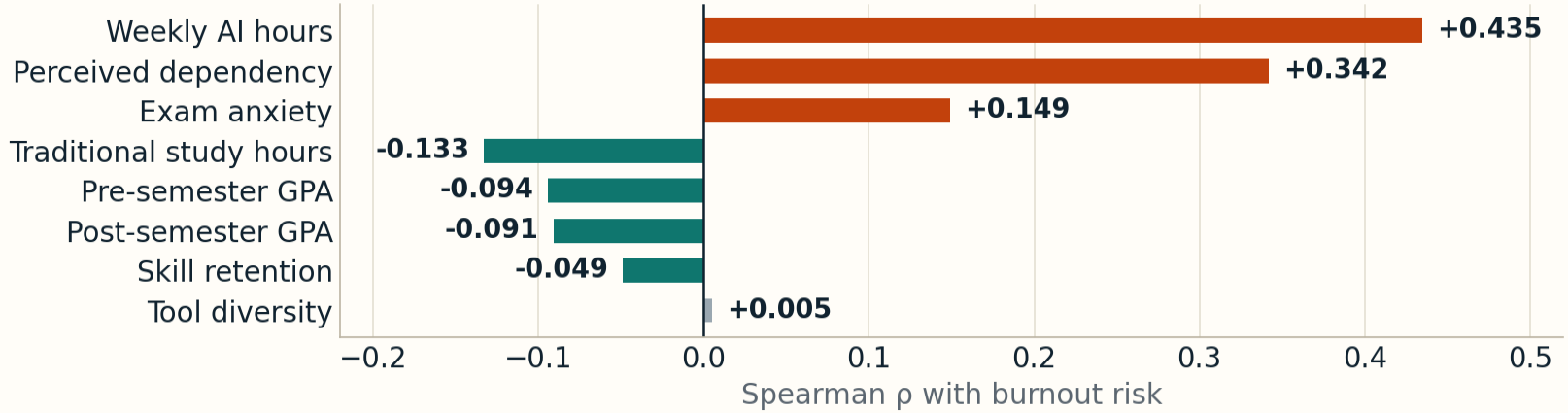
> Weekly hours is the most non-normal measure ($K^2 = 14{,}386.57$): a long right skew and a small cluster at the 40-hour ceiling.

Prior GPA dwarfs every AI variable



Weak direct links fit the hypothesis of non-linear, moderated relationships.

AI hours and dependency lead the burnout signal



Orange raises risk, teal protects; protects; offline study is the protective factor. factor.

TOOL DIVERSITY

More platforms do not mean more strain

ρ = +0.005

with burnout risk, p = 0.264 (not significant)

r = +0.025

with post-semester GPA

Negligible interface load

Cost of learning interfaces is small next to other factors.

Self-selection

Technically competent students may use more tools.

Complementary strengths

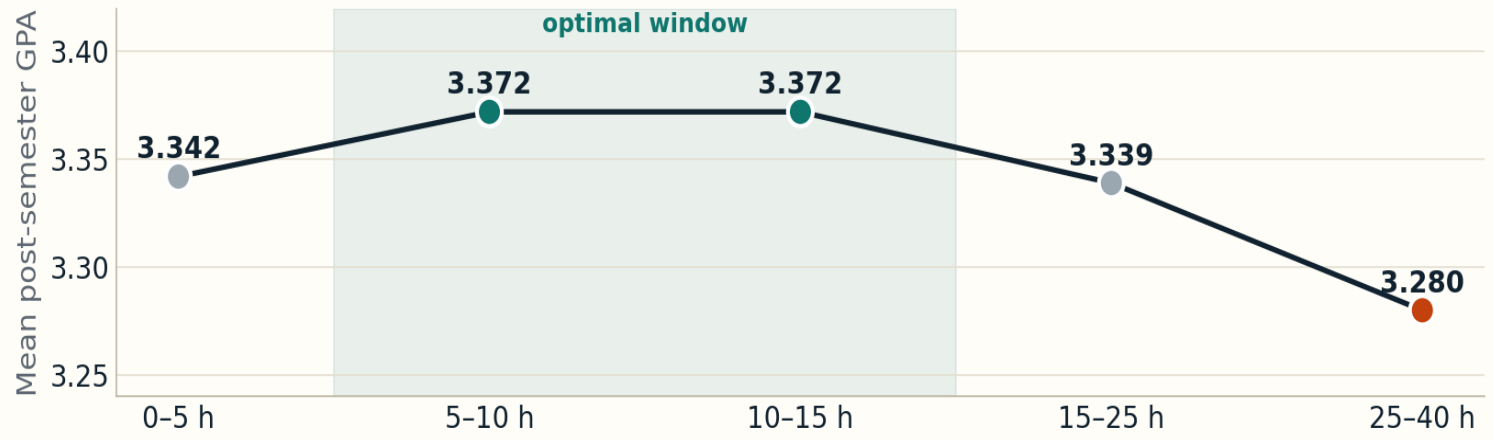
Different tools offset added complexity.

NON-LINEAR RELATIONSHIPS

Grades peak at 5–10 hours a week while burnout keeps climbing.

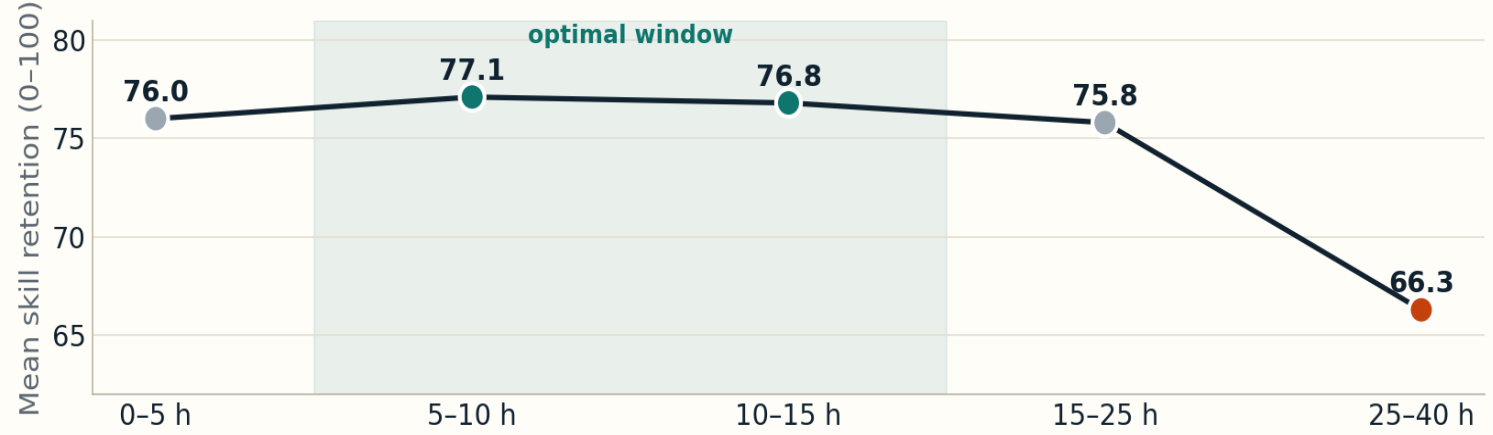
Usage bands reveal an inverted U for performance and an exponential climb for strain.

GPA is flat to 15 hours, then falls



Y-axis starts at 3.24 to make small differences visible. Decline is concentrated beyond 15 hours.

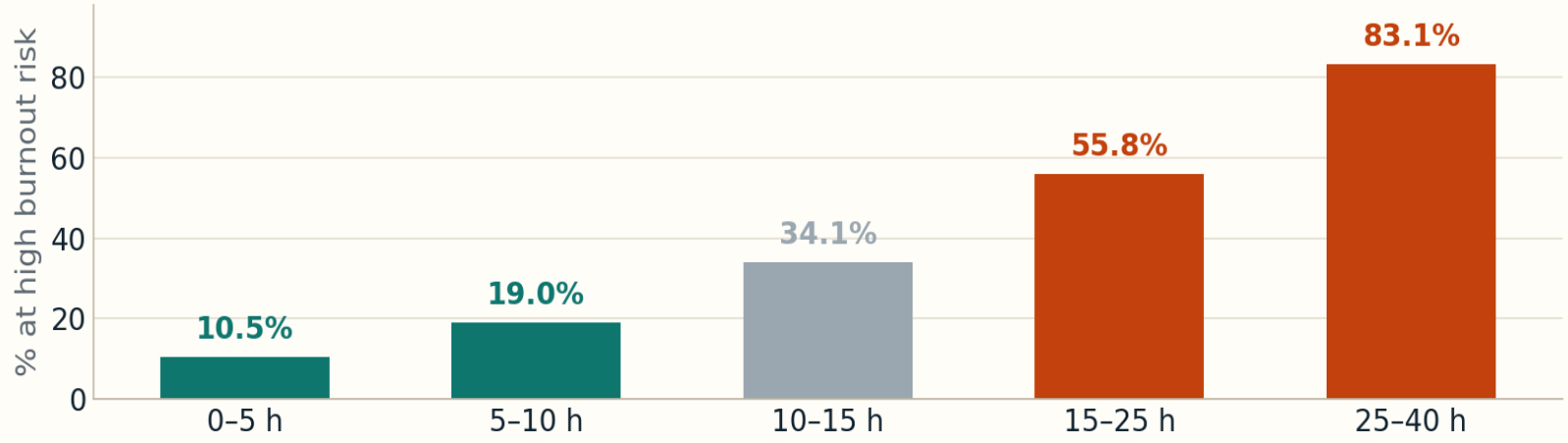
Retention collapses only in the heaviest band



Retention holds at 76–77 through 15 hours, then drops to 66.3 (y-axis starts starts at 62).

USAGE BANDS · BURNOUT

High burnout rises from 10.5% to 83.1%



Share of students at high burnout risk by weekly AI hours.

All three outcomes across five bands

Weekly AI hours	n	Post GPA	Skill retention	High burnout
0–5	22,592	3.342	76.0	10.5%
5–10	12,173	3.372	77.1	19.0%
10–15	6,701	3.372	76.8	34.1%
15–25	5,717	3.339	75.8	55.8%
25–40	2,817	3.280	66.3	83.1%

REGRESSION EVIDENCE . USAGE ZONES

Polynomial and piecewise models confirm the inverted U

+0.045
linear usage term, $p < 0.001$

−0.002
quadratic usage² term, $p < 0.001$

0–15 hours
Positive slope toward the optimum.

15–25 hours
Steep negative slope as risk escalates.

25–40 hours
Continued decline into a crisis point.

Three zones for practical guidance

5–15 h
Optimal window
Best GPA return; about 19% high burnout in the 5–10 band

15–25 h
Caution
Burnout climbs to 55.8%; gains shrink

25+ h
Crisis
83.1% high burnout; retention 66.3

> The institutional guidance bands later recommended: optimal 5–15, caution 15–25, crisis 25 and above.

DEPENDENCY THRESHOLD

Outcomes change regime above 6 out of 10

Perceived dependency	n	GPA	Skill retention	High burnout
Low (1–3)	13,000	3.35	76.8%	10.2%
Moderate (4–6)	21,500	3.36	76.2%	20.1%
High (7–10)	15,500	2.95	62.3%	94.7%

Threshold effect: $\chi^2 = 8{,}547$, $p < 0.001$ comparing groups below and above 6/10.

The cost of crossing 6 out of 10

−0.41
GPA points versus the moderate group
3.36 to 2.95

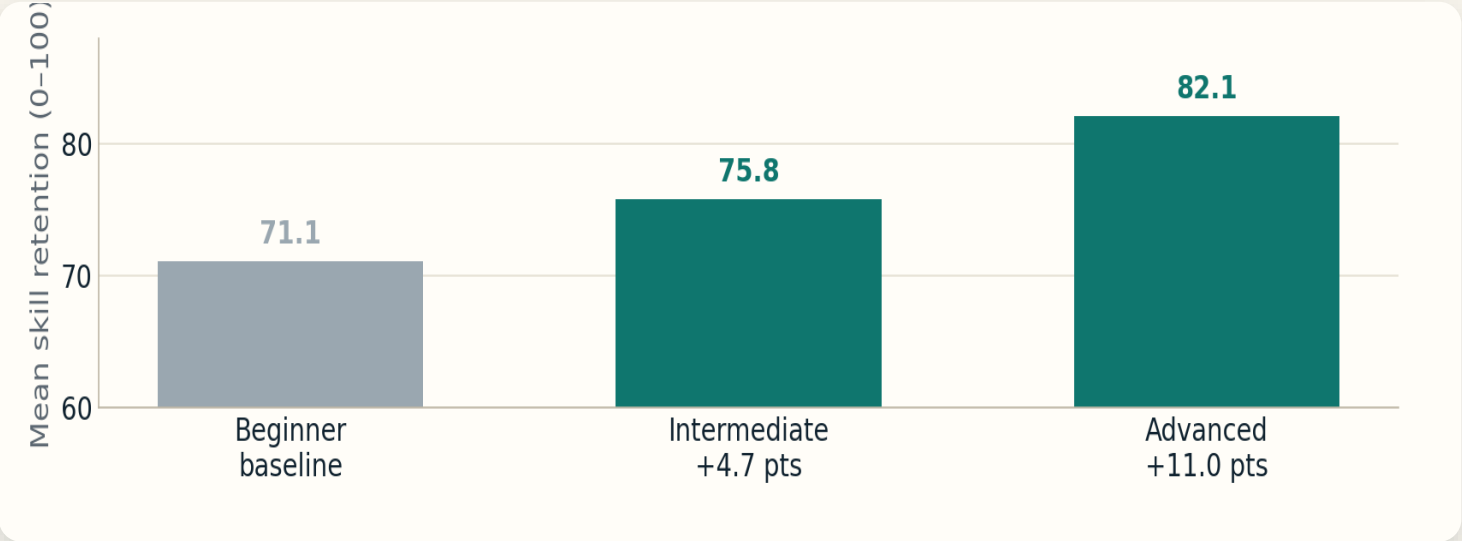
−14
skill-retention points
76.2% to 62.3%

+74.6
percentage points of high burnout
20.1% to 94.7%

Helping students keep a sense of autonomy may matter more than reducing hours alone.

PROMPT ENGINEERING SKILL

Advanced users retain 11 more points



The advanced edge is +15.5% relative to beginners (y-axis starts at 60).

Table 2: outcomes by proficiency

Proficiency	n (%)	Mean GPA	Retention	Gain vs. beginner
Beginner	16,750 (33.5%)	3.334	71.1	Baseline
Intermediate	16,750 (33.5%)	3.334	75.8	+4.7 pts (+6.6%)
Advanced	16,500 (33.0%)	3.389	82.1	+11.0 pts (+15.5%)

A large retention effect with almost no grade effect

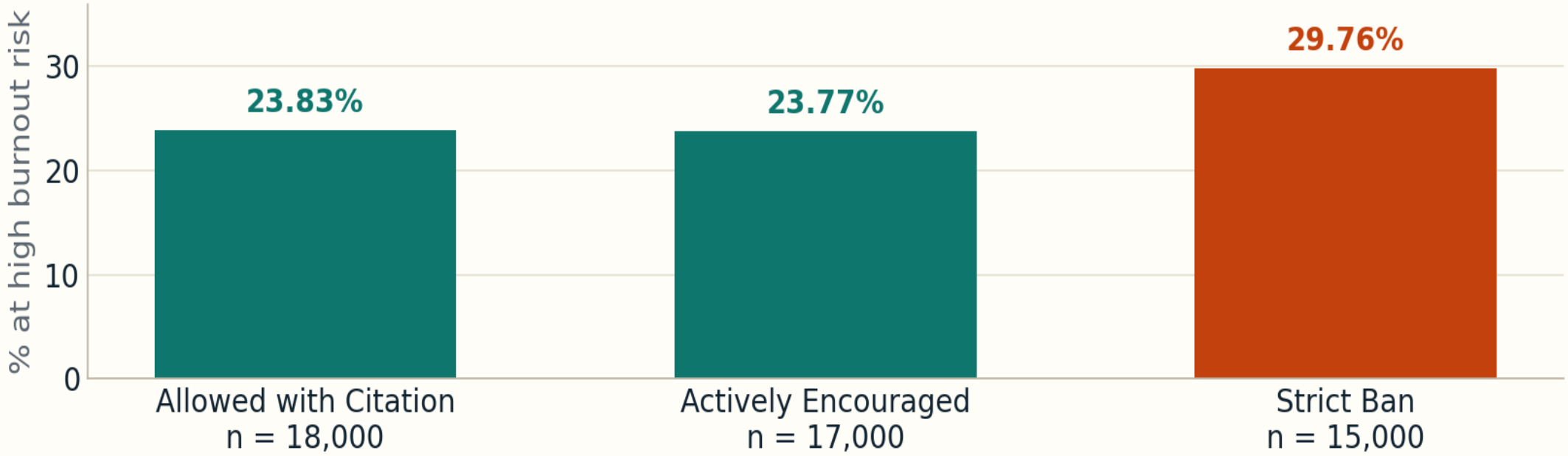
d = 0.84
effect size on skill retention (large)

d = 0.08
effect size on GPA (small, non-significant)

Prompt skill works as a learning multiplier rather than a shortcut: depth of learning changes, grades barely move.

INSTITUTIONAL POLICY

Strict bans carry the highest burnout



+5.93 points, a 25% relative increase under strict bans ($F = 127.4$, $p < 0.001$).

Table 3: outcomes by AI policy

Policy	n	Mean GPA	Retention	High burnout	Dependency
Allowed with Citation	18,000	3.35	75.91	23.83%	3.51
Actively Encouraged	17,000	3.35	76.13	23.77%	3.52
Strict Ban	15,000	3.33	74.99	29.76%	3.48

The two permissive frameworks are indistinguishable; the ban departs from both on burnout and, slightly, on GPA.

Prohibition does not reduce dependency

3.48 vs 3.51–3.52

perceived dependency: ban versus permissive

F = 0.87, p = 0.421

–0.021

GPA points under a strict ban

Tukey adjusted p = 0.0038 and 0.0014; ANOVA F = 6.834

Bans raise strain without changing the behaviour they target.

PREDICTION MODELS

Table 4: grades are predictable, burnout and retention are not

Task	Model	Primary metric
GPA (R^2)	Gradient Boosting	0.8871
GPA (R^2)	Random Forest	0.8845
GPA (R^2)	Linear Regression	0.8819
Skill retention (R^2)	Gradient Boosting	0.1531
Skill retention (R^2)	Random Forest	0.1488
Skill retention (R^2)	Linear Regression	0.0563
Burnout (accuracy)	Logistic Regression	0.5143
Burnout (accuracy)	Random Forest	0.5135

GPA MODEL · FEATURE IMPORTANCE

A strong model with limited actionability

0.887
R^2 for Gradient Boosting

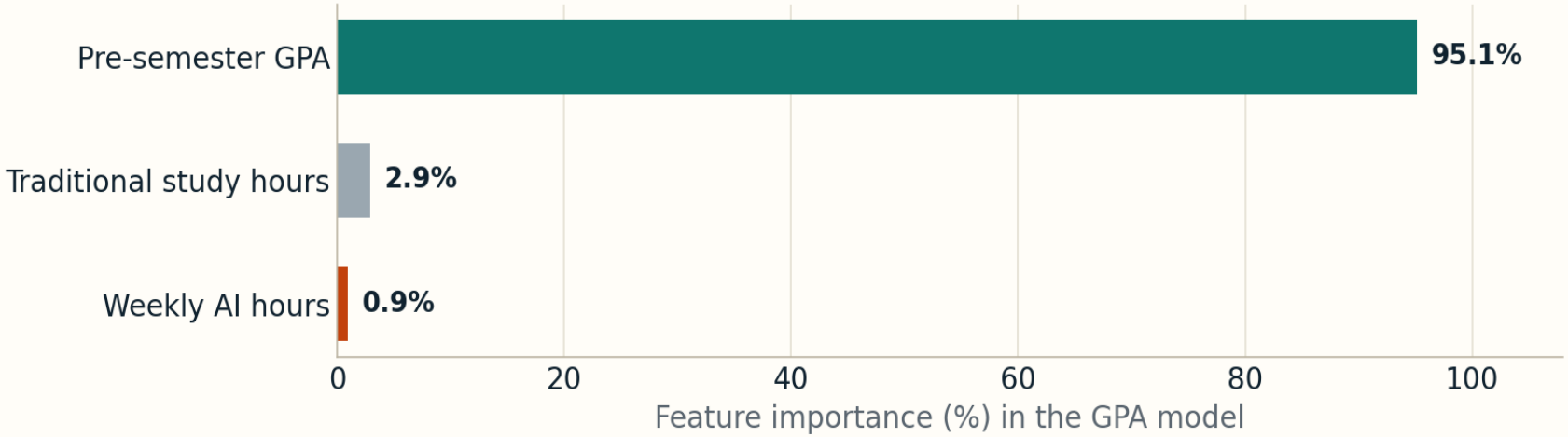
0.165
RMSE in GPA points

0.127
mean absolute error

99.2%
of predictions within ±0.5 GPA

Three algorithms differ by only 0.0026 in R^2, so a simple linear model serves forecasting about as well.

Pre-semester GPA carries 95.1% of the signal



Strong AI–GPA associations may reflect selection and confounding rather than causal effects of the tools.

Better than chance, not good enough for individuals

51.4%
accuracy, logistic regression
baseline 33.3%

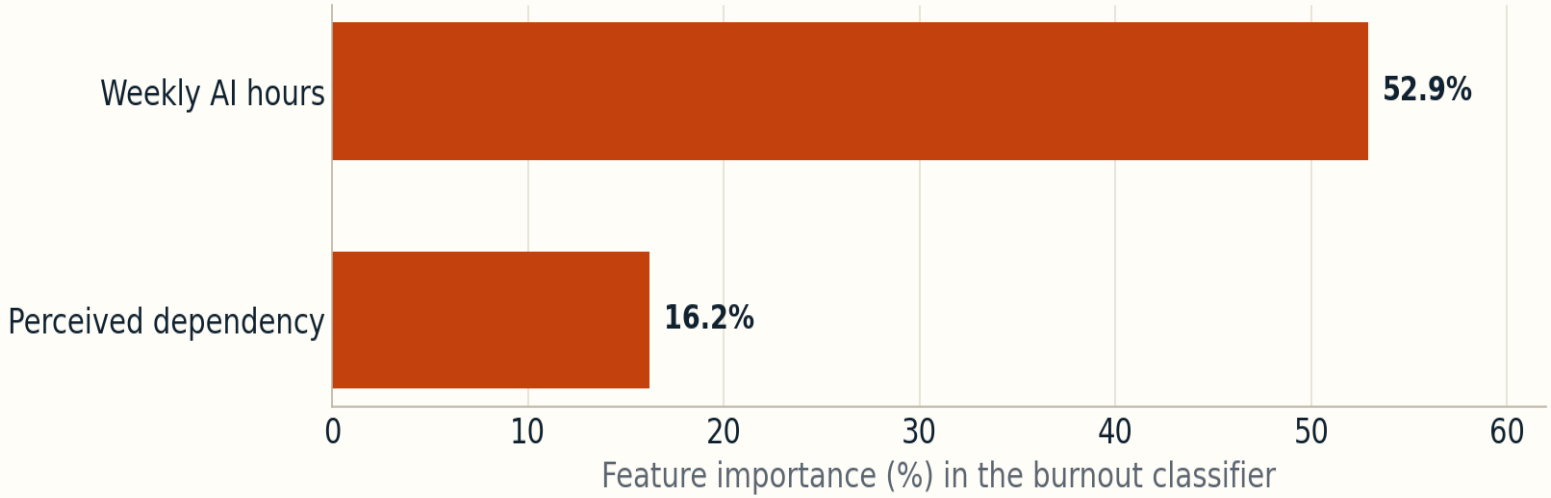
0.514
weighted F1 score

0.535
precision

0.514
recall

> Use as a screening aid that prompts a conversation or a validated instrument, not as a decision tool.

Usage hours and dependency dominate



Combined AI-related importance is 69.1%, yet accuracy stays weak: burnout depends on factors absent from the data.

Medium risk is the hardest class

Actual class	Predicted Low	Predicted Medium	Predicted High
Low	1,086	259	1,191
Medium	83	1,546	1,640
High	429	1,255	2,511

Medium students were labelled High (50.2%) more often than correctly identified (47.3%). High-risk recall is 59.9%.

Most of the variance remains unexplained

0.153
R^2 for Gradient Boosting

9.9
mean absolute error, in points

30.6%
predictions within ±5 points

57.0%
predictions within ±10 points

Retention is driven by factors outside this dataset, so the model is unfit for student-level decisions.

Different outcomes, different drivers

TASK A

GPA

Pre-semester GPA 95.1%
95.1%
Study hours 2.9%
AI hours 0.9%

TASK B

Burnout

AI hours 52.9%
Dependency 16.2%
Other predictors share the rest

TASK C

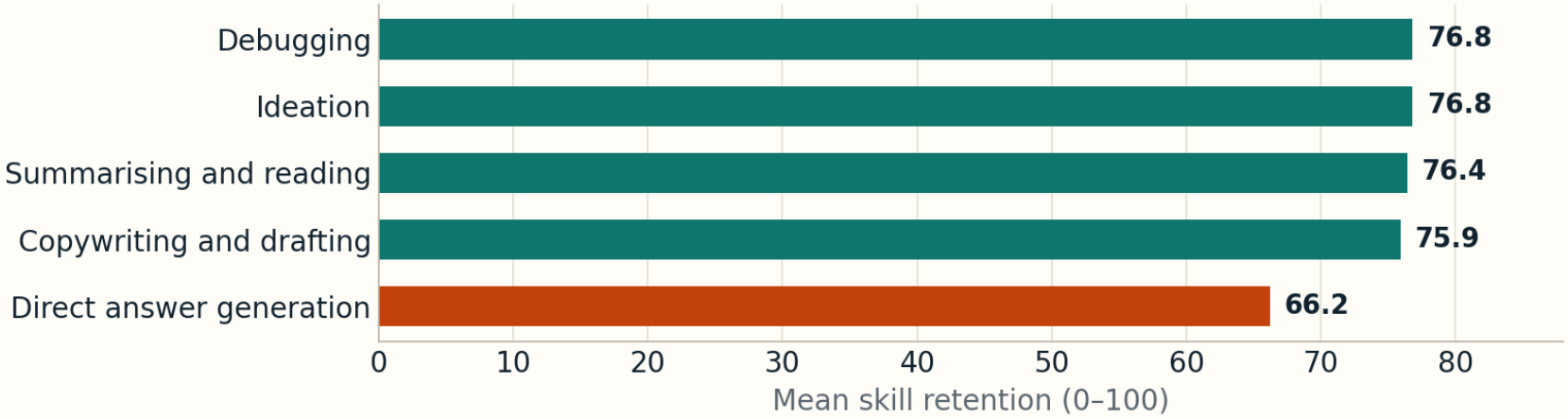
Skill retention
retention

Prompt skill 43.8%
AI hours 26.7%
Prior GPA 17.8%

AI variables add little for grades but but carry most of the signal for burnout and retention.

PRIMARY USE CASE

Direct answers cost 10 retention points



F = 247.3, p < 0.001. Debugging versus direct answers: +10.6 points, d = 0.76 (large).

Table 5: retention by use case

Primary use case	Students	n	Retention	vs. direct answers
Debugging	24.59%	12,295	76.8	+10.6 pts
Copywriting	24.02%	12,010	75.9	+9.7 pts
Ideation	21.44%	10,720	76.8	+10.6 pts
Summarising	17.27%	8,635	76.4	+10.2 pts
Direct answers	12.68%	6,340	66.2	Reference

DISCIPLINE

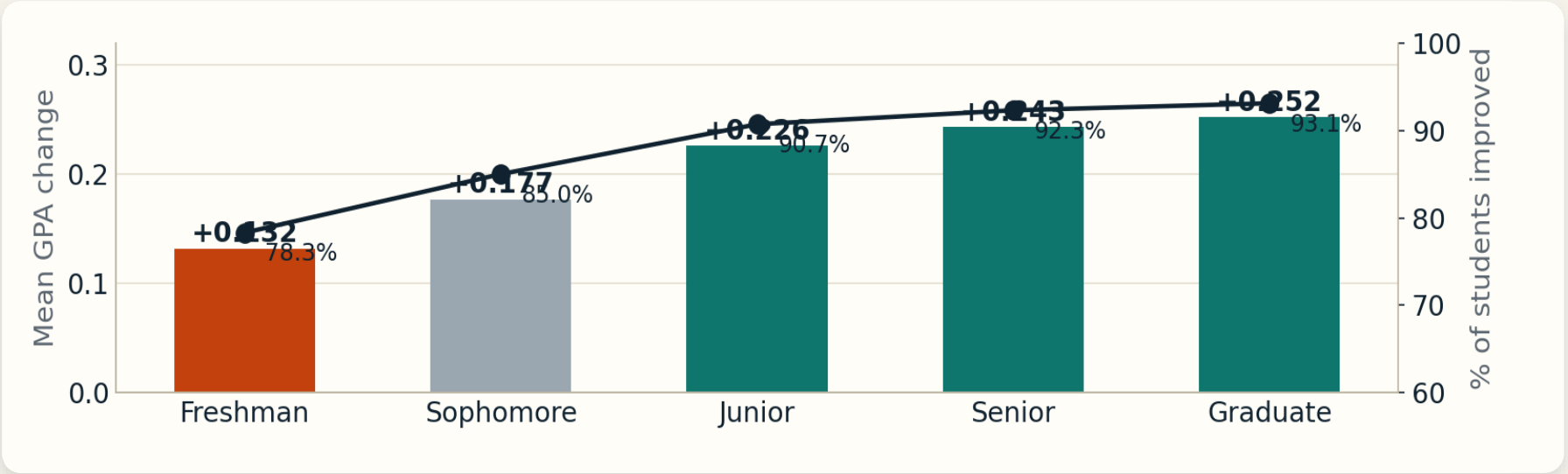
Gains are broadly shared across fields

Discipline	Share	GPA change	Skill retention
STEM	30.1%	+0.217	76.80
Business	25.1%	+0.194	75.26
Humanities	20.0%	+0.198	75.25
Medical	13.0%	+0.201	75.48
Arts	11.9%	+0.197	75.66

> Discipline effect on GPA change is negligible ($F = 1.14$, $p = 0.336$). High burnout: STEM 26.2% versus Humanities 22.1%.

YEAR OF STUDY

Gains grow with each year of study

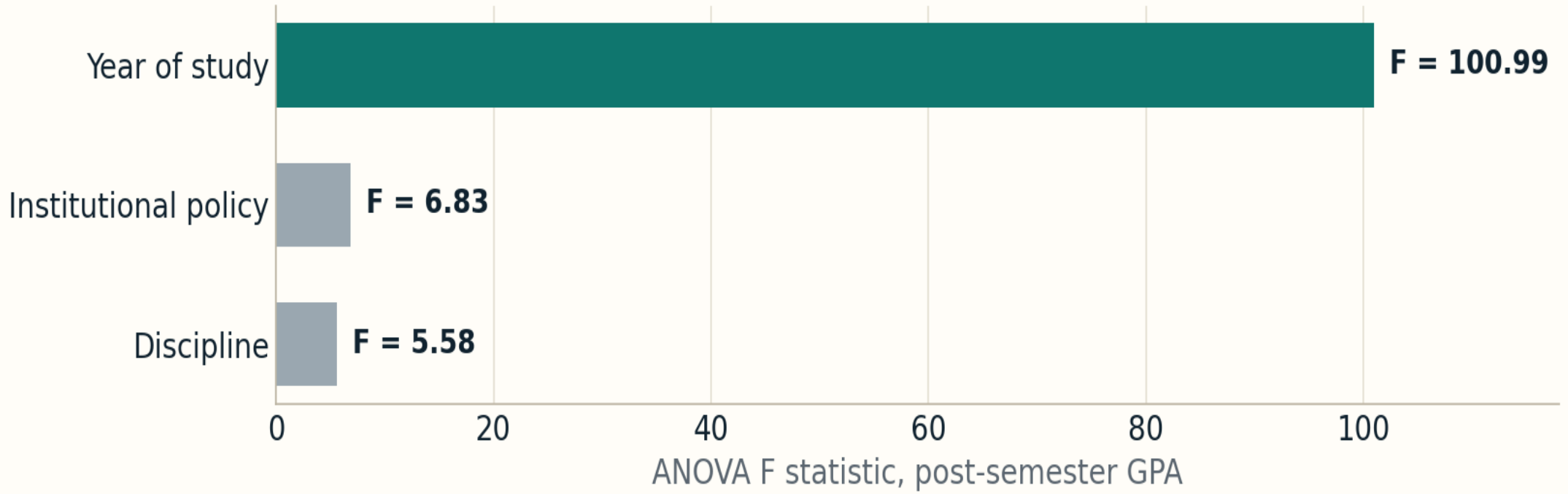


Pre-semester GPA was near identical across years (3.140–3.150).

Table 7: GPA outcomes by year

Year	n	Pre GPA	Post GPA	Change	Improved
Freshman	11,031	3.146	3.278	+0.132	78.3%
Sophomore	9,860	3.150	3.327	+0.177	85.0%
Junior	11,045	3.146	3.372	+0.226	90.7%
Senior	10,634	3.147	3.390	+0.243	92.3%
Graduate	7,430	3.140	3.392	+0.252	93.1%

Year of study is the largest between-group effect



Freshmen differ from every other year (all $p < 0.001$); seniors and graduates are indistinguishable ($p = 0.998$).

05

Discussion

What the patterns mean, why they may occur, and what to do about them.

PRINCIPAL FINDINGS

Five findings reshape the AI-in-education debate

1. **A non-linear usage window**
   Best returns at 5–15 hours a week; burnout escalates sharply past 20.

2. **Volume, not variety**
   Tool diversity is essentially unrelated to outcomes (r = −0.007).

3. **A dependency threshold**
   At 6/10 and above, retention falls from 76% to 62% and burnout surges from 20% to 95%.

4. **Prompt skill as multiplier**
   +11 retention points for advanced users with minimal GPA impact.

5. **Policy shapes strain**
   Strict bans yield the highest burnout and identical dependency.

HYPOTHESIS EVALUATION

Table 8: all seven hypotheses supported

	Prediction	Principal evidence	Assessment
H1	Inverted-U usage effect	✓ GPA 3.372 to 3.280; ✓ Burnout 10.5% to 83.1%	Supported
H2	Hours top burnout correlate	✓ ρ = +0.435; diversity ρ = +0.005	Supported
H3	Offline study protects	✓ ρ = −0.133	Supported, weak
H4	Dependency threshold	✓ Retention 62.3%, burnout 94.7%	Supported
H5	Skill moderates retention	✓ d = 0.84 versus GPA d = 0.08	Supported
H6	Bans raise burnout	✓ 23.83% versus 29.76%; ✓ dependency p = 0.421	Supported for burnout
H7	Use case moderates retention	✓ 76.8 versus 66.2, d = 0.76	Supported

Three processes likely sit behind the pattern

LEARNING

Cognitive

Removing productive struggle impairs retention, as seen for direct answers. Iterative prompting acts as cognitive apprenticeship.

AUTONOMY

Motivational

Excess assistance undermines autonomy; independent study may restore competence and autonomy, matching the offline-study buffer.

STRAIN

Occupational stress

Moderate use is optimal arousal. Past 15 hours, offloading and time pressure add strain.

Why hours and burnout may be linked

REVERSE

Coping

Burnout-prone students may use AI more as a last resort.

FORWARD

Fatigue

Heavy reliance may raise cognitive fatigue or lower agency.

CONFOUNDING

Stress-driven hours

Stressed students work longer and use AI proportionally more.

SELECTION

Self-selection

Capable students use more AI and also feel more efficacious.

RECOMMENDATIONS

Five evidence-based moves

1. **Publish usage guidance**
   Build it on the 5–15 hour window, for example 10–12 hours weekly.

2. **Make prompt literacy a core skill**
   Teach decomposition, task specification and iterative refinement.

3. **Adopt Allowed with Citation**
   Replace both prohibition and laissez-faire permissiveness.

4. **Redesign assessment**
   Reward reasoning depth: essays, oral defences, open-ended justification.

5. **Expand burnout support**
   Prioritise students with high dependency or 25+ weekly hours.

ACTION POINTS · STUDENTS AND EDUCATORS

Table 9,

Group	Do	Avoid
Students	Budget 10–12 AI hours weekly; favour debugging, ideation and summarising; build prompt skills; keep offline study at 10+ hours	More than 20 hours weekly; direct answer generation; ignoring a dependency score above 5/10
Educators	Teach explicit AI workflows; assess reasoning and process; include AI-free assessments; refer students with dependency signals	Treating AI as a policing problem; assignments a single prompt can solve

ACTION POINTS · INSTITUTIONS AND POLICYMAKERS

Group	Do	Avoid
Institutions	Adopt Allowed with Citation with published bands; fund AI literacy; ensure equitable access; integrate AI-use questions into mental-health services	Strict bans; leaving access to ability to pay (42.3% hold paid subscriptions)
Policymakers	Permit AI with disclosure; fund digital literacy as infrastructure; guarantee at least one high-quality tool; fund multi-year longitudinal research	One-size-fits-all mandates; ignoring equity effects of unequal access

Associations, not causes

Cross-sectional design

Direction of effects is unknown; reverse causation is plausible. Longitudinal tracking is needed.

Measurement validity

Burnout is categorical, retention was assessed one week after instruction, and dependency is self-reported.

What the data cannot see

Unmeasured variables

Conscientiousness, growth mindset, metacognition, instructor quality, life stressors and motivation.

Institutional context

Pooled institutions obscure differences in pedagogy, infrastructure and student populations.

Temporal specificity

One term, early in adoption. Patterns may shift as norms settle.

Confirms and extends earlier work

KULIK AND FLETCHER

Ed-tech effects

Earlier intelligent tutoring showed d = 0.40. Generative AI effects look more modest and depend on use case, skill and intensity.

WEI; ZAMFIRESCU-PEREIRA

Prompt competence

Provides quantitative support for skill as a moderator, with an 11-point retention edge rivalling many interventions.

BROD (1982)

Technostress

Tools can raise strain despite cutting objective workload, with non-linear escalation.

06

Conclusions

Six takeaways and the research agenda ahead.

CONCLUSIONS

Six takeaways

1

Usage window

Moderate use (5–15 hours) gives the best grades, about 19% high burnout.

2

Volume over variety

Hours predict burnout; tool diversity does not.

3

Prompt skill

+11 retention points with almost no GPA gap.

4

Dependency

At 6/10 or above retention is about 62% and burnout about 95%.

5

Policy

Citation beats bans on burnout with no dependency difference.

6

Use case

Scaffolding keeps retention at 76–77%; direct answers drop to 66%.

THE BOTTOM LINE

Pedagogical mastery, not automation, is the competitive advantage.

Guide intensity, teach prompt skill, replace bans with citation, and monitor dependency.

FUTURE RESEARCH

Four directions

CAUSALITY

Longitudinal designs

Track students across terms to establish causal order.

MEASURES

Richer psychology

Multidimensional burnout, self-efficacy, motivation and autonomy.

ACTION

Intervention trials

Test prompt training, usage caps and dependency monitoring.

DURABILITY

Durable transfer

Do retention gains carry to novel problems and later coursework?

COMPANION TOOL

AI Student Impact Analytics Dashboard

Dataset overview

Headline indicators, burnout distribution, composition and a filterable data table.

Advanced analytics

Performance, usage, burnout and retention views with ANOVA and Tukey tests.

ML models

GPA, burnout and retention models with comparison and diagnostics.

Predictions

Interactive predictions with risk status, confidence and tailored advice.

Insights

Critical findings and recommendations for four stakeholder groups.

Intended use

Exploratory and educational, not individual decisions about students.

QUESTIONS AND DISCUSSION

Thank you

Maurice Juma · October 2026

07

Appendices

Codebook, post-hoc comparisons and model diagnostics.

APPENDIX A · CODEBOOK

Table A1: variables in the dataset

Variable	Type	Values	Role
Student_ID	Identifier	Unique code	Record key
Major_Category	Categorical	STEM, Business, Humanities, Medical, Arts	Stratifier
Year_of_Study	Categorical	Freshman to Graduate	Stratifier
Pre_Semester_GPA	Continuous	1.0–4.0	Baseline
Post_Semester_GPA	Continuous	1.0–4.0	Primary outcome
Weekly_GenAI_Hours	Continuous	0–40 hours	Primary exposure
Traditional_Study_Hours	Continuous	0–40 hours	Protective factor
Tool_Diversity	Ordinal	1–5+ tools	Exposure

Variable	Type	Values	Role
Paid_Subscription	Binary	Free or Paid	Covariate
Primary_Use_Case	Categorical	Five use cases	Moderator
Prompt_Engineering_Skill	Ordinal	Beginner, Intermediate, Advanced	Moderator
Perceived_AI_Dependency	Ordinal	1–10	Threshold variable
Anxiety_Level_During_Exams	Ordinal	1–10	Predictor
Skill_Retention_Score	Continuous	0–100	Outcome
Burnout_Risk	Ordinal	Low, Medium, High	Outcome
Institutional_Policy	Categorical	Strict_Ban, Allowed_With_Citation, Actively_Encouraged	Moderator

Table B1, B2,& B3: post-semester GPA across disciplines, years of study and policies

Group 1	Group 2	Mean diff.	Adj. p	Sig.
Arts	Business	−0.0076	0.8672	
Arts	Humanities	0.0021	0.9989	
Arts	Medical	0.0095	0.8248	
Arts	STEM	0.0195	0.0774	
Business	Humanities	0.0097	0.5849	
Business	Medical	0.0171	0.1607	
Business	STEM	0.0271	0.0001	*
Humanities	Medical	0.0073	0.8858	
Humanities	STEM	0.0173	0.0525	
Medical	STEM	0.0100	0.6559	

Group 1	Group 2	Mean diff.	Adj. p	Sig.
Freshman	Graduate	0.1146	< 0.001	*
Freshman	Junior	0.0941	< 0.001	*
Freshman	Senior	0.1124	< 0.001	*
Freshman	Sophomore	0.0493	< 0.001	*
Graduate	Junior	−0.0204	0.0460	*
Graduate	Senior	−0.0022	0.9983	
Graduate	Sophomore	−0.0653	< 0.001	*
Junior	Senior	0.0182	0.0517	
Junior	Sophomore	−0.0449	< 0.001	*
Senior	Sophomore	−0.0631	< 0.001	*

Group 1	Group 2	Mean diff.	Adj. p	95% CI
Encouraged	Citation	0.0000	1.0000	[−0.0120, 0.0119]
Encouraged	Strict Ban	−0.0207	0.0038	[−0.0358, −0.0056]
Citation	Strict Ban	−0.0206	0.0014	[−0.0345, −0.0068]

One-way ANOVA on policy: $F = 6.834$, $p = 0.0011$.

Normality tests and prediction error

Variable	K^2 statistic	Decision
GPA difference	1,264.65	Reject normality
Weekly GenAI hours	14,386.57	Reject normality
Traditional study hours	411.56	Reject normality
Skill retention	500.58	Reject normality

Model	Median error	Max error	Within tolerance
GPA (Boosting)	0.103	1.004	±0.5: 99.2%; ±1.0: 100%
Retention (Boosting)	8.541 pts	42.197 pts	±5: 30.6%; ±10: 57.0%

Table C2: confusion matrices, test set n = 10,000

Logistic regression

Actual	Low	Medium	High
Low	1,086	259	1,191
Medium	83	1,546	1,640
High	429	1,255	2,511

Random forest

Actual	Low	Medium	High
Low	1,098	263	1,175
Medium	91	1,540	1,638
High	436	1,262	2,497